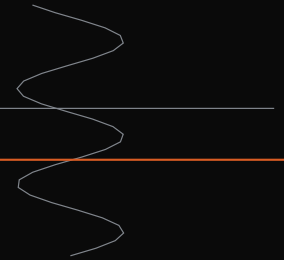
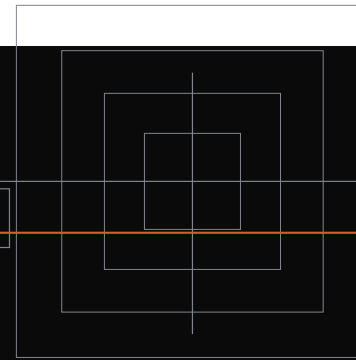


BILL OF MATERIALS

Robotic Arm (Arduino project)

Rev. 2026-04-30 49 items



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BOM

Product/Project: Robotic Arm (Arduino project)
Revision date: 2026-04-30

<https://www.printables.com/model/1439965-robotic-arm-arduino-project>

PRINTED PARTS

21 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	robot_belt_arm basering	1	AliExpress Link	Amazon Link	3D printed part #1 from the original Printables description.
>>	robot_belt_arm leg	x3	AliExpress Link	Amazon Link	3D printed part #2 from the original Printables description.
>>	robot_belt_arm socket_v2	1	AliExpress Link	Amazon Link	3D printed part #3 from the original Printables description.
>>	robot_belt_arm rotategear	1	AliExpress Link	Amazon Link	3D printed part #4 from the original Printables description.
>>	robot_belt_arm main_body_v2	1	AliExpress Link	Amazon Link	3D printed part #5 from the original Printables description.
>>	robot_belt_arm endstop	1	AliExpress Link	Amazon Link	3D printed part #6 from the original Printables description.
>>	robot_belt_arm gear_body	x2	AliExpress Link	Amazon Link	3D printed part #7 from the original Printables description.
>>	robot_belt_arm lever	1	AliExpress Link	Amazon Link	3D printed part #8 from the original Printables description.
>>	lower_shank_140	1	AliExpress Link	Amazon Link	3D printed part #9 from the original Printables description.
>>	upper_shank_140	1	AliExpress Link	Amazon Link	3D printed part #10 from the original Printables description.
>>	pleuel_140	1	AliExpress Link	Amazon Link	3D printed part #11 from the original Printables description.
>>	robot_belt_arm triplate	1	AliExpress Link	Amazon Link	3D printed part #12 from the original Printables description.

BOM	Product/Project: Robotic Arm (Arduino project) Revision date: 2026-04-30	https://www.printables.com/model/1439965-robotic-arm-arduino-project
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PRINTED PARTS continued					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	pleuel_bend_140	x2	AliExpress Link	Amazon Link	3D printed part #13 from the original Printables description.
>>	pleuel_bend_140 - mirrored	x2	AliExpress Link	Amazon Link	3D printed part #14 from the original Printables description.
>>	triplat_dual	1	AliExpress Link	Amazon Link	3D printed part #15 from the original Printables description.
>>	manipulator_dual	1	AliExpress Link	Amazon Link	3D printed part #16 from the original Printables description.
>>	gripperBase_by_ftobler	1	AliExpress Link	Amazon Link	3D printed part #17 from the original Printables description.
>>	gripper_shaft	1	AliExpress Link	Amazon Link	3D printed part #18 from the original Printables description.
>>	gripperFinger_by_ftobler	x2	AliExpress Link	Amazon Link	3D printed part #19 from the original Printables description.
>>	Wire_Guide	1	AliExpress Link	Amazon Link	3D printed part #20 from the original Printables description.
>>	robot_belt_arm stabilizer - myMOD	1	AliExpress Link	Amazon Link	3D printed part #21 from the original Printables description.

BOM

Product/Project: Robotic Arm (Arduino project)
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MECHATRONICS

16 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Nema17 Stepper Motor	x3	AliExpress Link	Amazon Link	Motor
>>	Arduino Mega 2560	1	AliExpress Link	Amazon Link	Controller board
>>	12V 4A Power Supply	1	AliExpress Link	Amazon Link	Power supply
>>	RAMPS 1.4	1	AliExpress Link	Amazon Link	Control shield / circuit board
>>	Female 2.1mm DC Jack	1	AliExpress Link	Amazon Link	DC barrel jack connector
>>	A4988	x3	AliExpress Link	Amazon Link	Stepper motor driver
>>	28BYJ-48	1	AliExpress Link	Amazon Link	12V gripper stepper motor
>>	ULN2003	1	AliExpress Link	Amazon Link	Driver board stepping module for 28BYJ-48
>>	End Stop	x3	AliExpress Link	Amazon Link	Limit switch for homing
>>	12V 5010 Fan	1	AliExpress Link	Amazon Link	Cooling fan
>>	2GT Pulley 20 Teeth	x3	AliExpress Link	Amazon Link	20-tooth 2GT pulley
>>	F624ZZ	x6	AliExpress Link	Amazon Link	Flange bearing
>>	F686ZZ	x12	AliExpress Link	Amazon Link	Bearing
>>	51105	1	AliExpress Link	Amazon Link	Ball thrust bearing
>>	2GT-6mm Closed Belt 200mm	x3	AliExpress Link	Amazon Link	Timing belt. Attached BOM image shows x3; the older description table listed x1.
>>	Paper clip (min. 35mm)	x2	AliExpress Link	Amazon Link	Gripper fingers link wire; present in the original description table, not shown in the attached image.

BOM

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FASTENERS

12 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	M2 10mm Button Head Screw	x6	AliExpress Link	Amazon Link	Button head screw M2x10mm. Attached BOM image shows x6; the older description table listed x10.
>>	M3 6mm Button Head Screw	x36	AliExpress Link	Amazon Link	Button head screw M3x6mm
>>	M4 10mm Button Head Screw	x6	AliExpress Link	Amazon Link	Button head screw M4x10mm
>>	M4 16mm Button Head Screw	x8	AliExpress Link	Amazon Link	Button head screw M4x16mm
>>	M4 20mm Button Head Screw	x6	AliExpress Link	Amazon Link	Button head screw M4x20mm
>>	M6 20mm Button Head Screw	1	AliExpress Link	Amazon Link	Button head screw M6x20mm
>>	M6 40mm Button Head Screw	x2	AliExpress Link	Amazon Link	Button head screw M6x40mm
>>	M6 50mm Button Head Screw	1	AliExpress Link	Amazon Link	Button head screw M6x50mm
>>	M2 Nut	x6	AliExpress Link	Amazon Link	Hexagonal nut M2
>>	M3 Nut	x4	AliExpress Link	Amazon Link	Hexagonal nut M3
>>	M4 Lock Nut	x14	AliExpress Link	Amazon Link	Hexagonal lock nut M4
>>	M6 Lock Nut	x4	AliExpress Link	Amazon Link	Hexagonal lock nut M6